

Lecture 5

nonlinear equations, definitions
bisection method, Picard's method
Newton-Raphson method

nonlinear equations, definitions

Example: nonlinear algebraic equation

The friction coefficient w for a fluid with particles satisfies the equation:

$$\frac{1}{\sqrt{w}} = \frac{\ln(\text{Re}\sqrt{w}) + 14 - \frac{5.6}{k}}{k},$$

where Re is the Reynolds number and k is an experimentally determined parameter.

To find w we need to solve this nonlinear algebraic equation. In this Lecture 9 we shall learn three methods: bisection, Picard's iteration, Newton-Raphson method.

It is always possible to rewrite the nonlinear equation in the form:

$$f(x) = 0$$

For example, the problem:

$$\text{find } w \text{ such that } \frac{1}{\sqrt{w}} = \frac{\ln(\operatorname{Re}\sqrt{w}) + 14 - \frac{5.6}{k}}{k},$$

can also be posed as:

$$f(x) = \frac{1}{\sqrt{x}} - \frac{\ln(\operatorname{Re}\sqrt{x}) + 14 - \frac{5.6}{k}}{k}, \quad \text{find } x \text{ such that } f(x) = 0$$

We say p is a *solution* of the equation $f(x) = 0$, if $f(p) = 0$ holds, i.e., we get exactly zero if we substitute the *number* p in place of the *variable* x .

Iterative solution procedure

- Suppose that there exists a solution p to the problem $f(x) = 0$.
- An *iterative* solution method generates a sequence of numbers:

$$p_0, p_1, \dots = \{p_n \mid n = 0, 1, \dots\}$$

- A method is considered successful if this sequence *converges* to p

Definition: convergence

The sequence $p_0, p_1, \dots$ *converges* to p if

$$\lim_{n \rightarrow \infty} p_n = p.$$

Alternative definitions:

$$\lim_{n \rightarrow \infty} (p - p_n) = 0 \quad \text{or} \quad \lim_{n \rightarrow \infty} |p - p_n| = 0$$

Theorem: condition for convergence

The sequence $p_0, p_1, \dots$ converges if and only if for any $\epsilon > 0$ there is a number N such that for all $n > N$ and $m > N$:

$$|p_n - p_m| < \epsilon$$

Theorem: one of the convergence criteria

Suppose that the sequence $p_0, p_1, \dots$ is such that, for some p ,

$$|p - p_n| \leq k|p - p_{n-1}|, \quad n = 1, 2, \dots$$

where $0 < k < 1$. Then, the sequence converges to p .

Proof: $|p - p_n| \leq k|p - p_{n-1}| \leq k^2|p - p_{n-2}| \leq \dots \leq k^n|p - p_0|$. Hence,

$$\lim_{n \rightarrow \infty} |p - p_n| = \lim_{n \rightarrow \infty} k^n |p - p_0| = |p - p_0| \lim_{n \rightarrow \infty} k^n = |p - p_0| (0) = 0$$



Definition: convergence order of the method

If an iterative method converges, $p_n \neq p$ for all n , and there exist $\alpha, \lambda > 0$ such that

$$\lim_{n \rightarrow \infty} \frac{|p - p_{n+1}|}{|p - p_n|^\alpha} = \lambda,$$

then the method has the *convergence order* α and the *asymptotic constant* λ .

- If $\alpha = 1$, then the method is *linearly convergent*, and λ is called the *asymptotic factor* in this case
- If $\alpha = 2$, then the method is *quadratically convergent*
- The higher the α , the faster the convergence

Stopping criteria

- Since we cannot wait for $n \rightarrow \infty$, we stop iterations at some finite n . When exactly to stop is decided by the *stopping criterion* – a condition such that one stops at the first n when this condition is satisfied and accepts p_n as an approximation for p .
- *Error criterion*: if p is known, choose some $\epsilon > 0$ and stop as soon as

$$|p - p_n| < \epsilon$$

- *Saturation/stagnation criterion*: if p is unknown, choose some $\epsilon > 0$ and stop as soon as the iterations start to saturate/stagnate:

$$|p_n - p_{n-1}| < \epsilon.$$

Residual

The *residual* of an equation $f(x) = 0$ at the step n of an iterative method is defined as:

$$r_n = f(p_n) - 0 = f(p_n)$$

Residual stopping criterion: if p is unknown, choose some ϵ and stop as soon as the magnitude of the residual is small enough:

$$|r_n| = |f(p_n)| < \epsilon$$

This should be an indicator of how close p_n is to p . But how good is it?

Connection between the residual and the error

We want to know the error $|p - p_n|$ but p is unknown. We can compute the residual $|f(p_n)|$ at every step. Let $f \in C^1[a, b]$ with $p \in [a, b]$. As soon as $p_n \in [a, b]$ we can use the Taylor series to write:

$$f(p_n) = f(p) + f'(\xi)(p - p_n), \quad \xi \in [a, b].$$

By definition of p , $f(p) = 0$. Therefore, the residual is directly proportional to the error:

$$|f(p_n)| = |f'(\xi)| |p - p_n|$$

If $f'(\xi) \neq 0$, and we use the residual criterion $|f(p_n)| < \epsilon$, then the corresponding error has the following bound:

$$|p - p_n| = \frac{|f(p_n)|}{|f'(\xi)|} \leq \frac{\epsilon}{m}, \quad m = \min_{\xi \in [a, b]} |f'(\xi)|$$

Ill-posed nonlinear problems

Since,

$$|p - p_n| = \frac{|f(p_n)|}{|f'(\xi)|},$$

the problem becomes ill-posed if $|f'(\xi)|$ is close to zero around or at the solution p .

A small perturbation in $f(x)$ can cause large errors in the approximation p_n , even if the residual $|f(p_n)|$ is very small.

Let's consider the question of perturbations in $f(x)$ more closely...

Uncertainty of solution in the presence of data errors

If the function $f(x)$ is not known exactly (experimental errors or numerical approximation errors), then we are dealing with an approximate $\hat{f}(x)$ instead of the exact $f(x)$.

Suppose, p is the solution of $f(x) = 0$ and $\hat{p}$ is the solution of $\hat{f}(x) = 0$, i.e., $\hat{f}(\hat{p}) = 0$. The true solution p can be anywhere in the *uncertainty interval*:

$$p \in [\hat{p} - \delta, \hat{p} + \delta], \quad \text{or} \quad |p - \hat{p}| \leq \delta.$$

To estimate δ , assume that $|\hat{f}(x) - f(x)| \leq \tilde{\epsilon}$ and use the Taylor series:

$$f(\hat{p}) = f(p) + f'(\xi)(p - \hat{p}), \quad \xi \in [a, b]$$

$$f(\hat{p}) = f'(\xi)(p - \hat{p}),$$

$$|f(\hat{p})| = |\hat{f}(\hat{p}) - f(\hat{p})| = |f'(\xi)| |p - \hat{p}| \leq \tilde{\epsilon},$$

$$|p - \hat{p}| \leq \frac{\tilde{\epsilon}}{|f'(\xi)|} = \delta$$

Uncertainty and stopping criterion

The uncertainty interval is:

$$|p - \hat{p}| \leq \frac{\tilde{\epsilon}}{|f'(\xi)|}, \quad \xi \in [a, b]$$
$$p \in \left[\hat{p} - \frac{\tilde{\epsilon}}{|f'(\xi)|}, \hat{p} + \frac{\tilde{\epsilon}}{|f'(\xi)|} \right].$$

Hence, for $\tilde{\epsilon} > 0$, the difference between $\hat{p}$ and p is unavoidable and it does not make sense to compute the exact solution $\hat{p}$ of the perturbed problem, i.e., to look for $\hat{p}$ such that $\hat{f}(\hat{p}) = 0$. Instead, the *residual* stopping criterion should be set as

$$|\hat{f}(p_n)| \leq \tilde{\epsilon}$$

Summary of theoretical facts

- A nonlinear problem can always be written in the form $f(x) = 0$
- Iterative methods of solving $f(x)$ generate sequences $p_0, p_1, \dots$ that may converge to the solution p
- Sequence converges if $\lim_{n \rightarrow \infty} p_n = p$
- While we want to stop an iterative method when the error stopping criterion $|p - p_n| < \epsilon$ is satisfied, in practice, either the residual $|f(p_n)| < \tilde{\epsilon}$ or the saturation/stagnation $|p_n - p_{n-1}| < \tilde{\epsilon}$ criteria are used
- Since most practical problems contain either experimental or truncation errors, it does not make sense to solve $\hat{f}(x) = 0$ exactly. This means that the stopping criterion for an iterative method should not contain a very small ϵ , and should be significantly above the machine precision ϵ_m .

bisection method

Idea of bisection method

The simplest method to find the solution of $f(x) = 0$ is based on the Intermediate Value Theorem. If $f \in C[a, b]$ and the sign of f is different in a and b , then 0 is an intermediate value of $f(x)$ and it is attained for some point $p \in [a, b]$.

The idea of bisection is:

- take the middle point of the interval $[a, b]$ as p_0
- from the two new sub-intervals $[a, p_0]$ and $[p_0, b]$ select the one where the sign at the boundaries changes and take the middle point of this sub-interval as p_1
- update your boundaries and continue until stopping criterion is met

Algorithm: Bisection Method

Let $f \in C[a, b]$ and $f(a)f(b) < 0$, then the *bisection method* produces the following iterations:

$$a_0 = a, \quad b_0 = b$$

for $n = 0, 1, \dots$ do:

$$p_n = \frac{a_n + b_n}{2}$$

if $f(a_n)f(p_n) < 0$:

$$a_{n+1} = a_n, \quad b_{n+1} = p_n$$

else :

$$a_{n+1} = p_n, \quad b_{n+1} = b_n$$

Theorem: Convergence of the Bisection Method

Let $f \in C[a, b]$ and $f(a)f(b) < 0$, then the *bisection method* produces a sequence $p_0, p_1, \dots$ that converges to the solution p of the equation $f(x) = 0$. Moreover,

$$|p - p_n| \leq \frac{b - a}{2^{n+1}}, \quad n \geq 0.$$

Proof: Since $b_n - a_n = (b - a)/2^n$, $p \in (a_n, b_n)$, and $p_n = (a_n + b_n)/2$, we have

$$|p - p_n| \leq \frac{b_n - a_n}{2} = \frac{b - a}{2^{n+1}}$$



Summary of Bisection Method

- Can be directly applied to problems of the form $f(x) = 0$
- Will always converge if the search interval $[a, b]$ is chosen correctly, i.e., if $f(a)f(b) < 0$
- ... order of convergence with the asymptotic factor $\lambda = \dots$ (see Questions on the last slide of this lecture)
- Stopping criterion of the form $\dots < \epsilon$ can be applied (see Questions on the last slide of this lecture)

Picard's method (fixed-point iterations)

Rewriting the problem in a fixed-point form

A fixed-point problem seeks x for which the following holds:

$$g(x) = x$$

Solution p of this problem is called a *fixed point*.

Any equation of the form $f(x) = 0$ can be written in a fixed-point form:

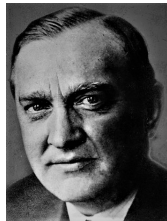
$$f(x) + x = x, \quad \text{or} \quad x - f(x) = x, \quad \text{or} \quad x - cf(x) = x$$

In some cases, other fixed-point forms are possible:

$$x^3 + 3x - 4 = 0, \quad \Rightarrow \quad \frac{4}{x^2 + 3} = x$$



Luitzen Egbertus Jan Brouwer (1881-1966)



Stefan Banach (1892-1945)

Theorem: Existence and uniqueness of a fixed point

Let $g \in C[a, b]$ and $g(x) \in [a, b]$ for all $x \in [a, b]$. Then, $g(x)$ has a fixed point in $[a, b]$ (Brouwer's Fixed-Point Theorem).

If, in addition, $g'(x)$ exists for all $x \in [a, b]$, and there exists $0 < k < 1$ such that

$$|g'(x)| \leq k, \quad \text{for all } x \in [a, b],$$

then the fixed point in $[a, b]$ is unique (Banach's Fixed-Point Theorem).

Proof:

Existence. If $g(a) = a$ or $g(b) = b$, then g has a fixed point in $[a, b]$. If $g(a) \neq a$ and $g(b) \neq b$, we look at the function $h(x) = g(x) - x$ and show that it equals zero for some $x \in [a, b]$. Since $g(x) \in [a, b]$, we have $g(a) - a = h(a) > 0$ and $g(b) - b = h(b) < 0$. Function $h(x)$ is continuous, because $g(x)$ is continuous. Hence, by the Intermediate Value Theorem, there is $x \in [a, b]$, where $h(x) = 0$.

Uniqueness. Suppose $|g'(x)| \leq k < 1$ and assume that there are two fixed points, p and q , in $[a, b]$, i.e., $g(p) = p$, $g(q) = q$, and $p \neq q$. By the Mean Value Theorem there exists $\xi \in [a, b]$ such that

$$\frac{g(p) - g(q)}{p - q} = g'(\xi).$$

Then,

$$|p - q| = |g(p) - g(q)| = |g'(\xi)| |p - q| < |p - q|,$$

which is a contradiction. Hence, $p = q$ and the fixed point is unique. □



Charles Émile Picard (1856-1941)

Algorithm: Picard's Method (fixed-point iterations)

Start with some *initial guess* p_0 and generate the following sequence:

$$p_n = g(p_{n-1}), \quad n = 1, 2, \dots$$

Remark. If $p_0, p_1, \dots$ converges to p , and $g \in C[a, b]$, then

$$p = \lim_{n \rightarrow \infty} p_n = \lim_{n \rightarrow \infty} g(p_{n-1}) = g(\lim_{n \rightarrow \infty} p_{n-1}) = g(p)$$

Meaning that the Picard's method will find the fixed point.

Convergence order of Picard's Method

Assume that there is a fixed point $p \in [a, b]$ (see Brouwer-Banach Theorem, Slide 21) and that the sequence $p_n = g(p_{n-1})$ converges to p . The order α and the asymptotic constant λ of an iterative method are defined via the relation (see Definition on Slide 7):

$$\lim_{n \rightarrow \infty} \frac{|p - p_{n+1}|}{|p - p_n|^\alpha} = \lambda$$

For Picard's Method with $0 < |g'(x)| \leq k$ for all $x \in [a, b]$, we get:

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{|p - p_{n+1}|}{|p - p_n|^\alpha} &= \lim_{n \rightarrow \infty} \frac{|g(p) - g(p_n)|}{|p - p_n|^\alpha} = \lim_{n \rightarrow \infty} \frac{|g'(\xi)| |p - p_n|}{|p - p_n|^\alpha} \\ &\leq k \lim_{n \rightarrow \infty} \frac{|p - p_n|}{|p - p_n|^\alpha} \end{aligned}$$

Hence, the method has *at least linear convergence*, $\alpha = 1$, in which case the asymptotic constant is $\lambda = k$. In general, it does not have a quadratic convergence, since the limit does not exist for $\alpha = 2$.

Stopping criterion for Picard's Method

Again, assume that there is a fixed point $p \in [a, b]$ and that the sequence $p_n = g(p_{n-1})$ converges to p . We want to stop the sequence when the error is small, i.e. $|p - p_n| < \epsilon$. However, we do not know p .

For convergent Picard's Method with $g'(x) \leq k$:

$$\begin{aligned} |p - p_n| &= \left| \lim_{m \rightarrow \infty} p_m - p_n \right| = \lim_{m \rightarrow \infty} |p_m - p_n| = \lim_{m \rightarrow \infty, m > n} |p_m - p_n| \\ &= \lim_{m \rightarrow \infty} |p_m - p_{m-1} + p_{m-1} - p_{m-2} - \cdots + p_{n+1} - p_n| \\ &\leq \lim_{m \rightarrow \infty} (|p_m - p_{m-1}| + |p_{m-1} - p_{m-2}| + \cdots + |p_{n+1} - p_n|) \end{aligned}$$

We note that:

$$\begin{aligned} |p_{n+1} - p_n| &= |g(p_n) - g(p_{n-1})| \leq k|p_n - p_{n-1}|, \\ |p_{n+2} - p_{n+1}| &= |g(p_{n+1}) - g(p_n)| \leq k|p_{n+1} - p_n| \leq k^2|p_n - p_{n-1}|, \\ &\quad \dots \\ |p_m - p_{m-1}| &\leq k^{m-n}|p_n - p_{n-1}| \end{aligned}$$

Stopping criterion for Picard's Method

Hence, with $k < 1$,

$$\begin{aligned}|p - p_n| &\leq \lim_{m \rightarrow \infty} (|p_m - p_{m-1}| + |p_{m-1} - p_{m-2}| + \cdots + |p_{n+1} - p_n|) \\&\leq \lim_{m \rightarrow \infty} (k^{m-n} |p_n - p_{n-1}| + k^{m-1-n} |p_n - p_{n-1}| + \cdots + k |p_n - p_{n-1}|) \\&= |p_n - p_{n-1}| \lim_{m \rightarrow \infty} (k^{m-n} + k^{m-1-n} + \cdots + k) \\&= k |p_n - p_{n-1}| \lim_{m \rightarrow \infty} (1 + k + \cdots + k^{m-n-1}) \\&= k |p_n - p_{n-1}| \lim_{r \rightarrow \infty} (1 + k + \cdots + k^r) \\&= \frac{k}{1-k} |p_n - p_{n-1}|,\end{aligned}$$

where we have recognized the Taylor series for $(1 - k)^{-1}$. Now, instead of the impractical error stopping criterion $|p - p_n| < \epsilon$, we can use the saturation/stagnation stopping criterion:

$$\begin{aligned}\frac{k}{1-k} |p_n - p_{n-1}| &< \epsilon \\|p_n - p_{n-1}| &< \frac{1-k}{k} \epsilon\end{aligned}$$

Theorem: Convergence of Picard's Method

Let $g \in C[a, b]$, $g(x) \in [a, b]$ and $|g'(x)| \leq k < 1$ for all $x \in [a, b]$. Then, the iterations

$$p_n = g(p_{n-1}), \quad n = 1, 2, \dots$$

converge to the unique fixed point $p \in [a, b]$ for any $p_0 \in [a, b]$.

Proof. Under the conditions of the Theorem there is a unique fixed point $p \in [a, b]$, where $g(p) = p$. Also, from the Mean-Value Theorem it follows:

$$|p - p_n| = |g(p) - g(p_{n-1})| = |g'(\xi)| |p - p_{n-1}| \leq k |p - p_{n-1}|,$$

where ξ is between p and p_{n-1} . Since $k < 1$, the sequence $p_0, p_1, \dots$ converges to p (second Theorem on Slide 6). The condition $p_0 \in [a, b]$ is required for $|g'(x)| \leq k < 1$ to hold at that point as well. □

Summary of Picard's Method

- Rewrite $f(x) = 0$ as a fixed-point problem $g(x) = x$
- Picard's iterations $p_n = g(p_{n-1})$, $n = 1, 2, \dots$
- Converges to the unique fixed point $p \in [a, b]$ for any $p_0 \in [a, b]$ if $g(x) \in [a, b]$ and $|g'(x)| \leq k < 1$ for all $x \in [a, b]$
- Method has at least linear convergence with the asymptotic factor $k < 1$
- Saturation/stagnation stopping criterion can be applied:

$$|p_n - p_{n-1}| < \frac{1-k}{k} \epsilon,$$

where ϵ is the desired error.

- Drawbacks: if $|g'(x)| \geq 1$ around $x = p$, the method will not converge; the domain $[a, b]$ where the method converges can be very small, i.e., the initial guess p_0 should be very close to p

Newton-Raphson method

Motivation of the Newton-Raphson method

Let the problem $f(x) = 0$ have the solution $x = p$. Expanding $f(p)$ around $x \neq p$, we get:

$$\begin{aligned}f(p) &= f(x) + f'(x)(p - x) + \frac{f''(\xi)}{2}(p - x)^2, \\0 &= f(x) + f'(x)(p - x) + \frac{f''(\xi)}{2}(p - x)^2,\end{aligned}$$

where ξ is between x and p . Suppose $f'(x) \neq 0$. Then,

$$\begin{aligned}p &= x - \frac{f(x)}{f'(x)} + \frac{f''(\xi)}{2f'(x)}(p - x)^2 \\p &\approx x - \frac{f(x)}{f'(x)}\end{aligned}$$

Algorithm: Newton-Raphson Method

Start with some initial guess p_0 and generate the sequence:

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad n = 1, 2, \dots$$

It is assumed that for any x the derivative $f'(x)$ is either available or can be approximated with a finite-difference formula

Theorem: Convergence of the Newton-Raphson Method

Let $f \in C^2[a, b]$. Suppose there is $p \in [a, b]$ where $f(p) = 0$ and $f'(p) \neq 0$. Then, there exists $\delta > 0$ such that the sequence generated by the Newton-Raphson Method converges to p for any $p_0 \in [p - \delta, p + \delta]$.

Proof.

The The Newton-Raphson Method

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad n = 1, 2, \dots$$

is, in fact, Picard's Method applied to the fixed-point problem $g(x) = x$ with

$$g(x) = x - \frac{f(x)}{f'(x)}.$$

From the Theorem on the Convergence of Picard's Method (Slide 28) we know that it will converge for any $p_0 \in [a, b]$ if:

- ① $g \in C[a, b]$
- ② $|g'(x)| \leq k < 1$ for all $x \in [a, b]$
- ③ $g(x) \in [a, b]$ for all $x \in [a, b]$

Proof.

- 1 Showing that there is an interval around p where $g(x)$ is continuous. We know that $f(x) \in C[a, b]$ and $f'(x) \in C[a, b]$, because $f \in C^2[a, b]$. The only danger to continuity of g is where $f'(x) = 0$. Since $f'(p) \neq 0$ and $f' \in C[a, b]$, there is $\delta_1 > 0$ such that $f'(x) \neq 0$ for all $x \in [p - \delta_1, p + \delta_1]$. Hence, $g \in C[p - \delta_1, p + \delta_1]$ for some $\delta_1 > 0$.
- 2 Showing that there is an interval around p where $|g'(x)| \leq k < 1$. The derivative of g is:

$$g'(x) = 1 - \frac{(f'(x))^2 - f(x)f''(x)}{(f'(x))^2} = \frac{f(x)f''(x)}{(f'(x))^2}$$

We see that $g' \in C[p - \delta_1, p + \delta_1]$ and $g'(p) = 0$. Hence, there is $\delta \leq \delta_1$, such that $|g'(x)| \leq k < 1$ for all $x \in [p - \delta, p + \delta]$.

- 3 Determining the range of $g(x)$ for $x \in [p - \delta, p + \delta]$. Let $x \in [p - \delta, p + \delta]$. From the Mean-Value Theorem:

$$\begin{aligned} |p - g(x)| &= |g(p) - g(x)| = |g'(\xi)||p - x| < |p - x| < \delta \\ -\delta < p - g(x) < \delta, &\Rightarrow g(x) \in [p - \delta, p + \delta] \end{aligned}$$

Order of convergence of the Newton-Raphson Method

Recall that the order of convergence α and the asymptotic constant λ are defined as those for which the following limit exists:

$$\lim_{n \rightarrow \infty} \frac{|p - p_{n+1}|}{|p - p_n|^\alpha} = \lambda$$

We do not have the explicit expression for $|p - p_n|$ yet. From the Taylor series of $f(p) = 0$ around p_n :

$$\begin{aligned} 0 = f(p) &= f(p_n) + f'(p_n)(p - p_n) + \frac{f''(\xi)}{2}(p - p_n)^2, \\ p - p_n &= -\frac{f(p_n)}{f'(p_n)} - \frac{f''(\xi)}{2f'(p_n)}(p - p_n)^2 \end{aligned}$$

where ξ is between p and p_n . From one step of the Newton-Raphson method:

$$\begin{aligned} p_{n+1} &= p_n - \frac{f(p_n)}{f'(p_n)} \\ p - p_{n+1} &= p - p_n - \frac{f(p_n)}{f'(p_n)} = \frac{f''(\xi)}{2f'(p_n)}(p - p_n)^2 \end{aligned}$$

Order of convergence of the Newton-Raphson Method

Substituting and taking into account that $\xi \rightarrow p$ as $n \rightarrow \infty$, we get:

$$\lim_{n \rightarrow \infty} \frac{|p - p_{n+1}|}{|p - p_n|^\alpha} = \lim_{n \rightarrow \infty} \frac{\left| \frac{f''(\xi)}{2f'(\xi)} (p - p_n)^2 \right|}{|p - p_n|^\alpha} = \left| \frac{f''(p)}{2f'(p)} \right| \lim_{n \rightarrow \infty} \frac{|p - p_n|^2}{|p - p_n|^\alpha}$$

The limit exists for at least $\alpha = 2$. Therefore, the Newton-Raphson Method has *quadratic order of convergence* with the asymptotic constant:

$$\lambda = \left| \frac{f''(p)}{2f'(p)} \right|$$

Stopping criterion for the Newton-Raphson Method

For iterative methods that work directly with $f(x)$, it is always possible to make a stopping criterion based on the residual $|f(p_n)|$. Say, we want to stop when the error $|p - p_n| < \epsilon$. Then, we can stop when:

$$|f(p_n)| = |f'(\xi)||p - p_n| \leq k|p - p_n| < k\epsilon,$$

where $0 < |f'(x)| \leq k$ for all $x \in [p - \delta, p + \delta]$. If you do not know k , simply make the residual as small as possible, remembering the role of experimental and numerical errors.

Summary of the Newton-Raphson Method

- Iterations $p_n = p_{n-1} - f(p_{n-1})/f'(p_{n-1})$, $n = 1, 2, \dots$
- Converges to the unique solution p for any $p_0 \in [p - \delta, p + \delta]$ if $f'(p) \neq 0$
- The domain of convergence $x \in [p - \delta, p + \delta]$ is where

$$\left| \frac{f(x)f''(x)}{(f'(x))^2} \right| \leq k < 1$$

- Method has at least quadratic convergence with the asymptotic factor

$$\lambda = \left| \frac{f''(p)}{2f'(p)} \right|$$

- Residual stopping criterion can be applied:

$$|f(p_n)| < |f'(\xi)|\epsilon,$$

where ϵ is the desired error and ξ is between p and p_n .

- Drawbacks: requires good initial guess p_0 and $f'(p_n)$ at each iteration

Questions

- How is the convergence order of an iterative method defined?
- What are the convergence order and the asymptotic constant of the Bisection Method? You can derive them by finding the (largest) α for which the limit in the Definition on Slide 7 exists. Hint: use the Theorem from Slide 18.
- Propose a stopping criterion for the Bisection Method. Hint: see Theorem on Slide 18